

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Java Spring Boot REST APIs** for inter-service communication, handling approximately 1,500 requests per minute during peak load.
- Developed and maintained **Hibernate/JPA persistence layers** for core data entities, ensuring data integrity across PostgreSQL and Oracle databases.
- Authored comprehensive **JUnit and Mockito unit tests** for critical Java components, achieving over 85% code coverage and reducing production defects by 15%.
- Built and managed **Linux-based device controllers** deployed in real-world environments, ensuring stable operation under unreliable network conditions.
- Engineered **real-time networking infrastructure** and device-to-cloud communication systems, facilitating seamless data exchange for access control.
- Developed **event-driven backend infrastructure** using message queues to process and coordinate device events, enhancing system responsiveness.
- Containerized backend services using **Docker** and deployed them to a managed **Kubernetes cluster** on AWS, improving deployment consistency and scalability.

SKILLS

Languages – Java, Python, SQL, TypeScript, Rust

Frameworks/Libraries – Spring Boot, Hibernate, JPA, NestJS, Node.js

Databases – PostgreSQL, MySQL, Oracle

Cloud/Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes

Tools/Platforms – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Linux

Analytics/BI – Power BI, Tableau, DAX, R, pandas, scikit-learn

PROJECTS

Real-time Device Fleet Management System (HCL Technologies)

- Developed a **Java Spring Boot backend** to manage a fleet of 5,000+ Linux controllers, handling device registration, status updates, and command dispatch.
- Implemented **distributed coordination mechanisms** using message queues to ensure reliable device state synchronization and fleet-wide command execution.
- Built **observability tooling** with Prometheus and Grafana to monitor system health, device connectivity, and performance metrics across the entire fleet.

Secure Access Control Backend with Device Identity (HCL Technologies)

- Engineered a **secure device authentication system** using TLS certificates and short-lived tokens, ensuring secure communication between devices and the cloud platform.
- Developed **REST APIs** with **FastAPI** for device management and user access control, integrating with an existing PostgreSQL database.
- Implemented **Python scripts** for automated certificate rotation and security policy enforcement across the device network.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)